

#if statement

a=23

b=200

if b>a:

print ("b is greater than a")

#elif

a=200

b=20

if b>a:

print("b is not greater than a")

else:

print("b is not greater than a")

#else

a=200

b=20

if b>a:

print("b is greater than a")

elif a==b:

print("a and b are equal")

else:

print("a is greater than b")

amount=2000

if amount<1000:

discount=amount*0.05

print("Discount",discount)

elif amount<5000:

discount=amount*0.10

```

    print("Discount",discount)
else:
    discount=amount*0.15
    print("Discount",discount)
    print("Net payable:",amount-discount)
#python nested if statement
var=100
if var<200:
    print("Expression value is less than 200")
    if var==150:
        print("Which is 150")
    elif var==100:
        print("Which is 100")
    elif var==50:
        print("Which is 50")
    elif var<50:
        print("expression value is less than 50")
    else:
        print("could not find true expression")
        print("Good bye!")
var=185
if var<200:
    print("Expression value is less than 200")
    if (var>150 and var<170):
        print("Which is greater than 150 less than 170")
    elif (var==100 or var>180):
        print("Which is equal to 100 or greater than 180")

```

```
elif var==50:

    print("Which is 50")

elif var<50:

    print("Expression value is less than 50")

else:

    print("could not find true expression")

    print("Good bye!")
```

#while loop statement

```
count=0

while(count<9):

    print('The count is:',count)

    count=count+1

print("Good bye!")
```

```
count=0

while count<7:

    print(count,"is less than 7")

    count=count+1

else:

    print("Loop end")
```

#for loop statement

```
range(0,5)

[0,1,2,3,4]

for var in list(range(5)):

    print(var)

for i in range(0,30,3):

    print(i)
```

#Break statement

```
for i in range(1,6):
```

```
    if i==3:
```

```
        break
```

```
    print(i)
```

```
#continue statement
```

```
for i in range(1,6):
```

```
    if i==3:
```

```
        continue
```

```
    print(i)
```

```
for i in range(1,4):
```

```
    pass
```

```
print("loop completed")
```

```
#empty condition block
```

```
x=10
```

```
if x>5:
```

```
    pass
```

```
else:
```

```
    print("Less than 5")
```

```
print("Done")
```

```
#old string formate
```

```
name="john"
```

```
print("Hello,%s!"%name)
```

```
name="john"
```

```
age=23
```

```
print("%s is %d years old."%(name,age))
```

```
#New string format{ }method
```

```

name="john"

age=23

print("My name is {} and i am {} years old.".format(name,age))

#python string format()method{:
txt1="My name is {fname},I'am {age}".format(fname="john",age=36)

txt2="My name is {0},I'am {1}".format("john",36)

txt3="My name is {},I'am {}".format("john",36)


print(txt1)

print(txt2)

print(txt3)

#python format

str="java"

str2="C#"

str3("{} and {} both are programming languages".format(str,str2)

print(str3)

#python program to do arithmetical operation

num1=input('Enter first number:')

num2=input('Enter second number:')

sum=float(num1)+float(num2)

min=float(num1)-float(num2)

mul=float(num1)*float(num2)

div=float(num1)/float(num2)

print('The sum of {0} and {1} is {2}'.format(num1,num2,sum))

print('The sumtraction of {0} and {1} is {2}'.format(num1,num2,min))

print('The multiplication of {0} and {1} is {2}'.format(num1,num2,mul))

```

```
print('The division of {0} and {1} is {2}'.format(num1,num2,div))
```

```
#python program to swap two variable
```

```
x=input('Enter value of x:')
```

```
y=input('Enter value of y:')
```

```
temp=x
```

```
x=y
```

```
y=temp
```

```
print('The value of x after swapping:{}'.format(x))
```

```
print('The value of y after swapping:{}'.format(y))
```

```
num=int(input("Enter a number:"))
```

```
if(num%2)==0:
```

```
    print("{} is Even number".format(num))
```

```
else:
```

```
    print("{} is Odd number".format(num))
```

```
output:
```

```
b is greater than a
```

```
b is not greater than a
```

```
a is greater than b
```

```
Discount 200.0
```

```
Expression value is less than 200
```

```
Which is 100
```

```
Expression value is less than 200
```

```
Which is equal to 100 or greater than 180
```

```
The count is: 0
```

```
The count is: 1
```

```
The count is: 2
```

```
The count is: 3
```

The count is: 4

The count is: 5

The count is: 6

The count is: 7

The count is: 8

Good bye!

0 is less than 7

1 is less than 7

2 is less than 7

3 is less than 7

4 is less than 7

5 is less than 7

6 is less than 7

Loop end

0

1

2

3

4

0

3

6

9

12

15

18

21

24

27

1

2

1

2

4

5

loop completed

Done

Hello,john!

john is 23 years old.

My name is john and i am 23 years old.

My name is john,I'am 36

My name is john,I'am 36

My name is john,I'am 36

java and C# both are programming languages

Enter first number:2

Enter second number:10

The sum of 2 and 10 is 12.0

The sumtraction of 2 and 10 is -8.0

The multiplication of 2 and 10 is 20.0

The division of 2 and 10 id 0.2

Enter value of x:3

Enter value of y:5

The value of x after swapping:5

The value of y after swapping:3

Enter a number:2

2 is Even number